Intro to Web Dev

Assignment - 01

Q1. Explain the difference between frontend, backend, and full-stack development with suitable real-world examples.

Ans . 1. Frontend Development

Frontend development deals with the visible and interactive part of a website or web application that users directly interact with.

Technologies: HTML, CSS, JavaScript, React, etc.

Example: On an e-commerce website, the product cards, search bar, navigation menu, buttons, and shopping cart interface are part of the frontend.

2. Backend Development

Backend development deals with the server-side logic and operations that happen behind the website. It handles requests, authentication, business logic, and communication with databases.

Technologies: Node.js, Python, Java, PHP, etc.

Example: When a user logs into an e-commerce website, the backend verifies the username and password and retrieves the user's account information from the database.

3. Full-Stack Development

Full-stack development involves working on both frontend and backend, along with connecting the application to databases and APIs.

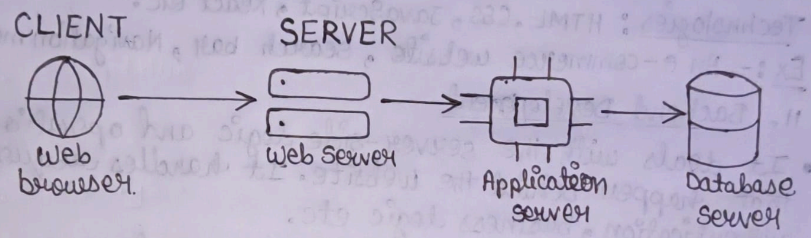
Example: A full-stack developer can build an entire e-commerce application—from designing the product page and shopping cart to creating the backend for orders, payments, user accounts, and database management.

Q2. Create a simple diagram showing how the client-server model works in web architecture.

Ans.

Ans2 The client-server model is an architecture in which the client (web browser) sends a request to a server, and the server processes the request and sends back the required response.

Client - Server Model in Web Development



Basic Flow

Client (Browser) ⟶ HTTP/HTTPS Request ⟶ Web Server ⟶ Database (if required) ⟶ Web Server ⟶ HTTP/HTTPS Response ⟶ Client.

Q3. Describe how a browser requests and displays a web page from a web server.

Ans. When a user enters a website URL in a browser, the following steps take place:

1. **URL is entered:** The user enters a website address, such as www.example.com.
2. **DNS lookup:** The browser finds the IP address of the web server using DNS.
3. **Request is sent:** The browser sends an HTTP/HTTPS request to the web server.
4. **Server processes the request:** The server processes the request and, if required, retrieves data from a database.
5. **Response is sent:** The server sends an HTTP/HTTPS response containing HTML, CSS, JavaScript, images, and other resources.
6. **Browser renders the page:** The browser processes these resources and uses its rendering engine to create and display the webpage on the screen.

Q4. Identify and list the tools required to set up a web development environment. Explain the purpose of each.

Ans. The following tools are commonly required for a basic web development environment:

Tool	Purpose
Visual Studio Code (VS Code)	Used to write and edit HTML, CSS, JavaScript, and other code.
Web Browser	Used to run, test, and debug websites. Examples: Chrome, Firefox, Edge.
Git	Used for version control and tracking changes in the project.
GitHub	Used to store code online and collaborate with other developers.
Node.js	Provides a JavaScript runtime and is required by many modern web development tools.
npm	Used to install and manage JavaScript packages and dependencies.
Terminal / Command Prompt	Used to run commands, install packages, and start development servers.

Q5. Explain what a web server is and give examples of commonly used servers.

Ans. A web server is a software or computer system that receives requests from web browsers and sends back web resources such as HTML pages, CSS, JavaScript, images, and API responses.

When a user visits a website, the browser sends a request to the web server. The server processes the request and returns the required content to the browser.

Commonly Used Web Servers

1. **Apache HTTP Server** – A widely used open-source web server.
2. **Nginx** – Known for high performance and is commonly used for serving websites and as a reverse proxy.
3. **Microsoft IIS** – A web server developed by Microsoft for Windows servers.
4. **LiteSpeed** – A high-performance web server often used for hosting websites.
5. **Caddy** – A modern web server known for simple configuration and automatic HTTPS.

Example: When you open a website in your browser, the browser sends a request to its web server, which responds with the files required to display the webpage.

Q6. Define the roles of a frontend developer, backend developer, and database administrator in a project.

Ans. 1. Frontend Developer

A frontend developer is responsible for the user interface and user experience of a website or application.

Main responsibilities:

- Creates webpage layouts using HTML, CSS, and JavaScript.
- Makes the website responsive and interactive.
- Connects the interface with backend APIs.

2. Backend Developer

A backend developer is responsible for the server-side logic and functionality of the application.

Main responsibilities:

- Develops APIs and server-side logic.
- Handles authentication and authorization.
- Processes requests and communicates with databases.
- Implements the application's business logic.

3. Database Administrator (DBA)

A database administrator is responsible for managing and maintaining the database used by the application.

Main responsibilities:

- Creates and manages databases.
- Maintains data security and integrity.
- Performs backups and recovery.
- Monitors and improves database performance.

Q7. Install VS Code and configure it for HTML, CSS, and JavaScript development. Take a screenshot of the setup.

Ans. Steps to Set Up VS Code

1. Download and install VS Code on your computer.

2. Open VS Code and create a new folder for your web project.
3. Open the project folder in VS Code.
4. Create three files:
    - index.html – for HTML structure
    - style.css – for styling
    - script.js – for JavaScript functionality
5. Link the CSS and JavaScript files with the HTML file.

Q8. Explain the difference between static and dynamic websites. Provide an example of each.

Ans.

Static Website	**Dynamic Website**
Content is mostly fixed and changes only when the developer updates the files.	Content can change automatically based on users, data, or requests.
Mainly uses HTML, CSS, and JavaScript.	Usually uses frontend, backend, and database technologies.
Usually does not require a database.	Often uses a database to store and retrieve data.
Easier and faster to develop and host.	More complex to develop and maintain.
Suitable for portfolios, landing pages, and informational websites.	Suitable for e-commerce, social media, banking, and web applications.

Examples

Static Website:
A personal portfolio containing fixed sections such as **About, Skills, Projects, and Contact.**

Dynamic Website:
An e-commerce website where products, prices, user accounts, orders, and recommendations can change based on database data and user activity.

Q9. Research and list five web browsers. Explain how rendering engines differ between them.

Ans. A **rendering engine** is the component of a web browser that processes HTML, CSS, and other web resources and converts them into the webpage displayed on the screen.

Web Browser	**Rendering Engine**
Google Chrome	Blink
Microsoft Edge	Blink
Mozilla Firefox	Gecko
Apple Safari	WebKit
Opera	Blink

How do they differ?

Different rendering engines have different implementations and optimizations for processing HTML, CSS, and web standards**.** Therefore, a webpage may sometimes look or behave slightly differently across browsers.

Q10. Draw a labeled diagram showing the basic web architecture flow — client, server, database, and APIs.

Ans.